

Involvement of TRPV4 in temperature-dependent perspiration in mice

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Abstract

Reports indicate that an interaction between TRPV4 and anoctamin 1 could be widely involved in water efflux of exocrine glands, suggesting that the interaction could play a role in perspiration. In secretory cells of sweat glands present in mouse foot pads, TRPV4 clearly colocalized with cytokeratin 8, anoctamin 1 (ANO1) and aquaporin-5 (AQP5). Mouse sweat glands showed TRPV4-dependent cytosolic Ca^{2+} increases that was inhibited by menthol. Acetylcholine-stimulated sweating in foot pads was temperature-dependent in wild-type, but not in TRPV4-deficient mice, and was inhibited by menthol both in wild-type and TRPM8KO mice. The basal sweating without acetylcholine stimulation was inhibited by an ANO1 inhibitor. Sweating could be important for maintaining friction forces in mouse foot pads, and this possibility is supported by the finding that wild-type mice climbed up a slippery slope more easily than TRPV4-deleted mice. Furthermore, TRPV4 expression was significantly higher in controls and normohidrotic skin from patients with AIGA (acquired idiopathic generalized anhidrosis) compared to anhidrotic skin from patients with AIGA. Collectively, TRPV4 is likely involved in temperature-dependent perspiration via interactions with ANO1, and TRPV4 itself or the TRPV4/ANO 1 complex would be targets to develop agents that regulate perspiration.

eLife assessment

This **useful** studying implicates TRPV4 as a mediator of sweat, potentially based on TRPV4's expression and function on sweat glands. The data and methods are **solid**, with some limitations in terms of the approach. Overall, the work lends new insight into the physiologic basis of sweating using data from mice and humans.

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Introduction

Sweating is a vital physiological process (Shibasaki and Crandall 2010 [DOI](#)). There are two basic types of sweating: thermoregulatory and emotional sweating, in addition to gustatory sweating, largely localized to the face and neck regions, that occurs while consuming some foods, particularly pungent foods (Lee 1954 [DOI](#)). Most sweat glands are of the eccrine type, and they produce a thin secretion that is hypotonic to plasma. Although eccrine sweat glands are distributed all over the body, their density is highest in the axillary region and on the palms of the hands and the soles of the feet. In humans, the main function of eccrine sweat glands is body temperature regulation. Meanwhile, apocrine sweat glands are found primarily in the axillae and urogenital regions. These scent glands become active during puberty and secrete a viscous fluid that is associated with body odor.

Body temperature regulation is important to maintain a homeostasis. Body temperature is poorly controlled in patients with hypohidrosis. Meanwhile, patients affected by hyperhidrosis can have difficulty in social and professional situations due to increased sweat production, and the resulting subjective perception of illness at an individual level may be substantial (Cohen and Solish 2003 [DOI](#), Schlereth, Dieterich et al. 2009 [DOI](#)). However, the molecular mechanisms of perspiration are not clearly understood.

We previously reported the functional interaction between TRP channels and the Ca^{2+} -activated Cl^- channel, anoctamin 1 (ANO1, also known as TMEM16A) (Takayama, Shibasaki et al. 2014 [DOI](#), Takayama, Uta et al. 2015 [DOI](#), Derouiche, Takayama et al. 2018 [DOI](#)). TRP channels have high Ca^{2+} permeability (Gees, Owsianik et al. 2012 [DOI](#)) and Ca^{2+} entering cells through TRP channels activates ANO1 by making a physical complex, leading to Cl^- efflux in cells with high intracellular Cl^- concentrations. The Cl^- efflux may drive water efflux through water channels in exocrine gland acinar cells that increases exocrine function, and causes depolarization in primary sensory neuron that increases nociception. For skin keratinocytes that have relatively low intracellular Cl^- concentrations, interaction between TRPV3 and ANO1 causes Cl^- influx, followed by increased cellular movement/proliferation in response to cell cycle modulation (Yamanoi, Lei et al. 2023 [DOI](#)). Thus, direction of Cl^- movement through ANO1 is simply determined by the balance between equilibrium potentials of Cl^- and membrane potentials in each cell (Takayama, Derouiche et al. 2019 [DOI](#)).

Involvement of TRPV4 in exocrine gland function prompted us to examine the functional interaction in perspiration, because TRPV4 is expressed in human eccrine sweat glands (Delany, Hurle et al. 2001 [DOI](#)). Although sweat glands are innervated by sympathetic neurons, acetylcholine is released from the nerve endings (Hu, Converse et al. 2018 [DOI](#)). We show that the functional interaction of TRPV4 and ANO1 is involved in temperature-dependent sweating and increased friction force.

Results

Expression of TRPV4, anoctamin 1 and AQP5 in mouse sweat glands

We detected expression of TRPV4, ANO1 and the water channel aquaporin-5 (AQP5) in eccrine glands of mouse foot pads. The secretory coil is located in the deep dermis and a relatively straight duct opens to the skin surface. We first validated an anti-TRPV4 antibody that we generated. This anti-TRPV4 antibody conspicuously labeled the basal layer of the epidermis, secretory eccrine gland cells, and duct cells only in skin from wild-type (WT) mice, but not in skin from TRPV4-

deficient (TRPV4KO) mice (**Figure 1A** [↗](#)), indicating the antibody specificity. TRPV4 was clearly localized in secretory glands as confirmed by positivity for cytokeratin 8 (CK8), a secretory cell marker (**Figure 1B** [↗](#)). The duct cells were not labeled by ANO1 and CK8 (**Figure 1B** [↗](#)). TRPV4-immunoreactivity was stronger in duct cells near the secretory region and gradually diminished in the distal excretory ducts toward the epidermis. Bilayered sweat ducts showed TRPV4 labeling in basal cells but not suprabasal cells (**Figure 1C** [↗](#)). Secretory cells in human eccrine glands are classified into two types: clear cells that mainly secrete water and electrolytes, and dark cells that secrete macromolecules like glycoproteins. We found that TRPV4-expressing secretory cells were positive for the calcitonin gene-related peptide (CGRP), a dark cell marker, and were heterogeneously labeled (**Figure 1D** [↗](#)). This result is consistent with earlier studies showing that mouse eccrine glands have a more primitive structure than human glands, and have only one type of secretory cell that resembles human clear cells but also has dark cell characteristics (Kurosumi and Kurosumi 1970 [↗](#), Bovell 2018 [↗](#)).

To explore TRPV4 subcellular localization, we observed tissues using Airyscan super-resolution imaging. TRPV4 was heterogeneously labeled in the gland cells, and showed apparent localization in basal and apical membranes (**Figure 1D** [↗](#)). TRPV4 was absent in myoepithelial cells. Conspicuous co-labeling of TRPV4 and ANO1 or AQP5 with filamentous actin (F-actin) was seen at the apical site (luminal side) of the secretory cells (**Figure 1E** [↗](#)). These close topological relationships clearly suggest that TRPV4, ANO1 and AQP5 would be able to form a complex that promotes sweat secretion in eccrine glands of mouse foot pads. These results also suggest that TRPV4-expressing secretory cells are involved in secretion of macromolecular components as well as secretion of water and ions.

Functional expression of TRPV4 in acinar cells of mouse sweat glands

We previously showed the functional interaction of TRPV4 and ANO1 in the heterologous expression with HEK293T cells and mouse native exocrine gland acinar cells (Takayama, Shibasaki et al. 2014 [↗](#), Derouiche, Takayama et al. 2018 [↗](#)). Then, we examined functional TRPV4 expression in sweat glands in mouse foot pads. WT mouse sweat glands responded to the TRPV4 agonist, GSK (500 nM) and to acetylcholine (Ach, 10 μ M) (**Figure 2A** [↗](#)). No cytosolic Ca^{2+} increase induced by GSK was observed in sweat glands from TRPV4KO mice (**Figure 2B** [↗](#)). Interestingly, the GSK-induced increase in cytosolic Ca^{2+} was significantly inhibited by menthol (5 μ M) in WT mouse sweat glands, suggesting that menthol inhibited TRPV4 function as we reported before (Takayama, Furue et al. 2017 [↗](#)). Meanwhile, menthol alone caused no change in cytosolic Ca^{2+} concentration (**Figure 2C** [↗](#)). These data indicated functional expression of TRPV4 in mouse secretory cells.

TRPV4 involvement in perspiration in mice

To examine the functional interaction between TRPV4 and ANO1 in mouse sweat glands *in vivo*, stimulated sweating induced by Ach (100 μ M, 2 min) in mouse hind paws at 25 °C and 35 °C was investigated using an iodine and starch reaction to measure secreted amylase (Nejsum, Kwon et al. 2002 [↗](#)). At 25 °C, no difference in stimulated sweating was seen both in WT and TRPV4KO mice while sweating was increased at 35 °C in WT, but not in TRPV4KO mice (**Figure 3A** [↗](#)). Temperature-dependent basal sweating without Ach stimulation for 15 min was also observed for WT mice, but not in TRPV4KO mice (**Figure 3B** [↗](#)). Menthol, an well-known TRPM8 activator, inhibits both TRPV4 and ANO1 function (Takayama, Furue et al. 2017 [↗](#)). The ability of menthol to inhibit both TRPV4 and ANO1 suggests that menthol would inhibit sweating. Accordingly, we compared stimulated sweating with either ethanol vehicle (used for menthol dilution) or menthol treatment for 2 min. Menthol treatment caused a significantly lower degree of sweating than ethanol treatment both in WT and TRPM8KO mice (**Figure 3C, D** [↗](#)). This result could indicate that menthol inhibits sweating by inhibiting the function of both TRPV4 and ANO1, independently of

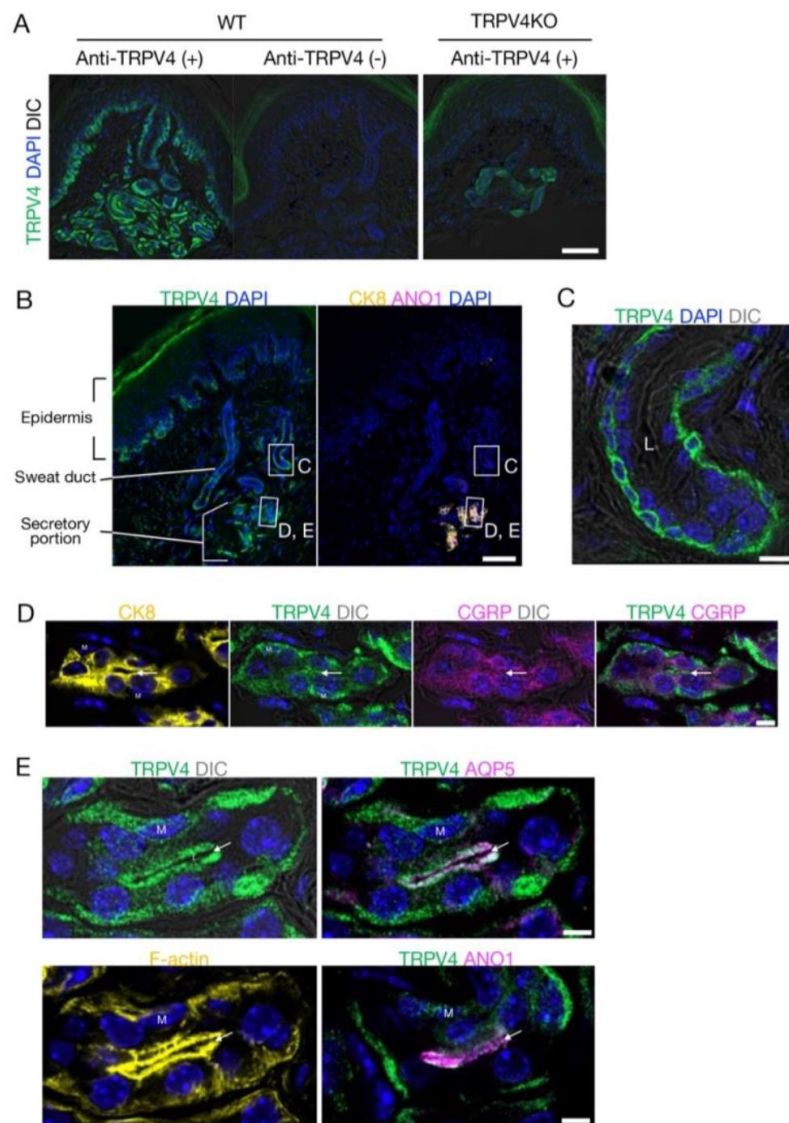


Figure 1.

TRPV4 localization in eccrine glands of mouse foot pads.

(A) TRPV4 signals in the secretory coil in the deep dermis with a relatively straight duct opening to the skin surface. (B) Localization of TRPV4 (green), cytokeratin 8 (CK8; yellow) and anoctamin 1 (ANO1; magenta) in the skin. (C, D, E) Highly magnified Airyscan super-resolution images of the sweat duct (C) and secretory portion (D, E). (C) TRPV4 localizes to the basal cells of the bilayered sweat duct. Ductal lumen: L. (D) Secretory gland showing labeling for TRPV4 and calcitonin gene-related peptide (CGRP). (E) Secretory gland with conspicuous TRPV4 labeling in myoepithelial cells (M) and secretory cells. TRPV4 clearly colocalizes with aquaporin-5 (AQP5), F-actin, and ANO1 at the luminal side of the secretory cells. Arrows indicate the glandular lumen. Differential interference contrast: DIC, Nuclei: DAPI. Scale bar: 50 μ m (A, B). 5 μ m (C-E).

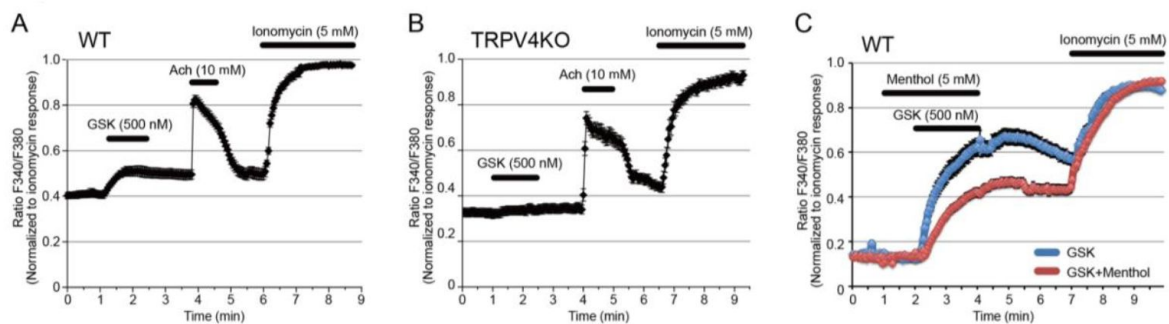


Figure 2.

Functional TRPV4 expression in mouse sweat gland acinar cells.

(A, B) Changes in cytosolic Ca^{2+} concentrations upon stimulation with GSK, acetylcholine or ionomycin in sweat gland acinar cells from wild-type (WT, A) and TRPV4-deficient (TRPV4, B) mice. $n=6$ experiments for WT and TRPV4KO sweat glands. (C) Changes in cytosolic Ca^{2+} concentration upon stimulation with GSK in the presence (red) or absence (blue) of menthol in sweat gland acinar cells from WT mice.

TRPM8. Next In order to prove the involvement of ANO1 in basal sweating, we examined the effects of a strong and specific ANO1 antagonist, Ani9. An Ani9 treatment almost completely abolished basal sweating both at 25 °C and 35 °C (**Figure 3E, F** [↗](#)), indicating a pivotal role of ANO1 in sweating.

Physiological significance of TRPV4-mediated sweating

Mice do not sweat to control body temperature, so the physiological significance of hind paw sweating is unclear. In humans, fingertip moisture is known to be optimally modulated during object manipulation through regulation of friction force ([Andre, Lefevre et al. 2010](#) [↗](#)). The same mechanism might promote for traction of hind paws when mice climb slippery slopes. Here we constructed a slope covered with slippery vinyl (**Figure 4A** [↗](#)) and compared climbing behaviors of WT and TRPV4KO mice for 1 hour at 26–27 °C with 35–50% humidity. The total number of climbing attempts was the same for WT and TRPV4KO mice (25.6 ± 2.5 for WT, $n = 5$; 24.7 ± 3.9 for TRPV4KO, $n = 4$) (**Figure 4B** [↗](#)), but a higher percentage of WT mice successfully climbed to the top of the slope than did TRPV4KO mice ($79.5 \pm 6.4\%$ for WT; $41.8 \pm 2.8\%$ for TRPV4KO; $p < 0.01$) (**Figure 4CD** [↗](#)) (Suppl. video). And WT mice easily came down the slippery slope. These data suggest that WT mice might produce more hind paw sweat (**Figure 3** [↗](#)) that increases traction on the slope.

TRPV4 expression in human sweat glands

We next examined whether TRPV4 also plays a role in human perspiration. Patients with acquired idiopathic generalized anhidrosis (AIGA) have acquired impairment in total body sweating even when exposed to heat or engaging in exercise ([Nakazato, Tamura et al. 2004](#) [↗](#), [Munetsugu, Fujimoto et al. 2017](#) [↗](#), [Sano, Asahina et al. 2017](#) [↗](#)). We compared TRPV4 expression in sweat glands from patients with melanocytic nevus ($n = 10$, ages; 15 – 63) as controls and patients with AIGA ($n = 10$, ages; 24 – 55) using a commercially-available anti-TRPV4 antibody different from the one utilized in the mouse samples. All patients with AIGA were male, which is consistent with the gender distribution of AIGA, while two of the 10 controls were female. Representative TRPV4 staining is shown in **Figure 5A, B** [↗](#). Although signals for TRPV4 staining were high in normohidrotic skin from a patient with AIGA and were equivalent to those of controls, levels in anhidrotic skin from the same patient with AIGA were very low.

We classified TRPV4 staining intensity from 1+ (low) to 3+ (high). Scores were significantly higher in controls and normohidrotic skin from patients with AIGA (2+ or 3+) than anhidrotic skin from AIGA cases (1+ or 2+) (mean 2.5 ± 0.17 vs 1.0 ± 0.10 for controls and normohidrotic skin from patients with AIGA vs. anhidrotic skin from AIGA cases, respectively, $p < 0.0001$) (**Figure 5** [↗](#)). These data clearly indicated that TRPV4 plays a role in normal perspiration in humans.

Discussion

Ca^{2+} entering cells through TRP channels is known to be involved in various Ca^{2+} -mediated events, particularly in non-excitable cells, whereas cation influx-induced depolarization is important for excitation of primary sensory neurons through activation of voltage-gated Na^+ channels ([Takayama, Shibasaki et al. 2014](#) [↗](#), [Takayama, Uta et al. 2015](#) [↗](#), [Derouiche, Takayama et al. 2018](#) [↗](#)). Ca^{2+} entering cells is instantaneously chelated to maintain low intracellular Ca^{2+} concentrations. However, high Ca^{2+} conditions can persist for longer periods just beneath the plasma membrane. We reported that several TRP channels including TRPV1, TRPV3, TRPV4 and TRPA1 can form a complex with the Ca^{2+} -activated Cl^- channel, ANO1, and activate ANO1 via Ca^{2+} entering cells through TRP channels ([Takayama, Shibasaki et al. 2014](#) [↗](#), [Takayama, Uta et al. 2015](#) [↗](#), [Derouiche, Takayama et al. 2018](#) [↗](#)) Interaction between TRPV4 and ANO1 causes Cl^-

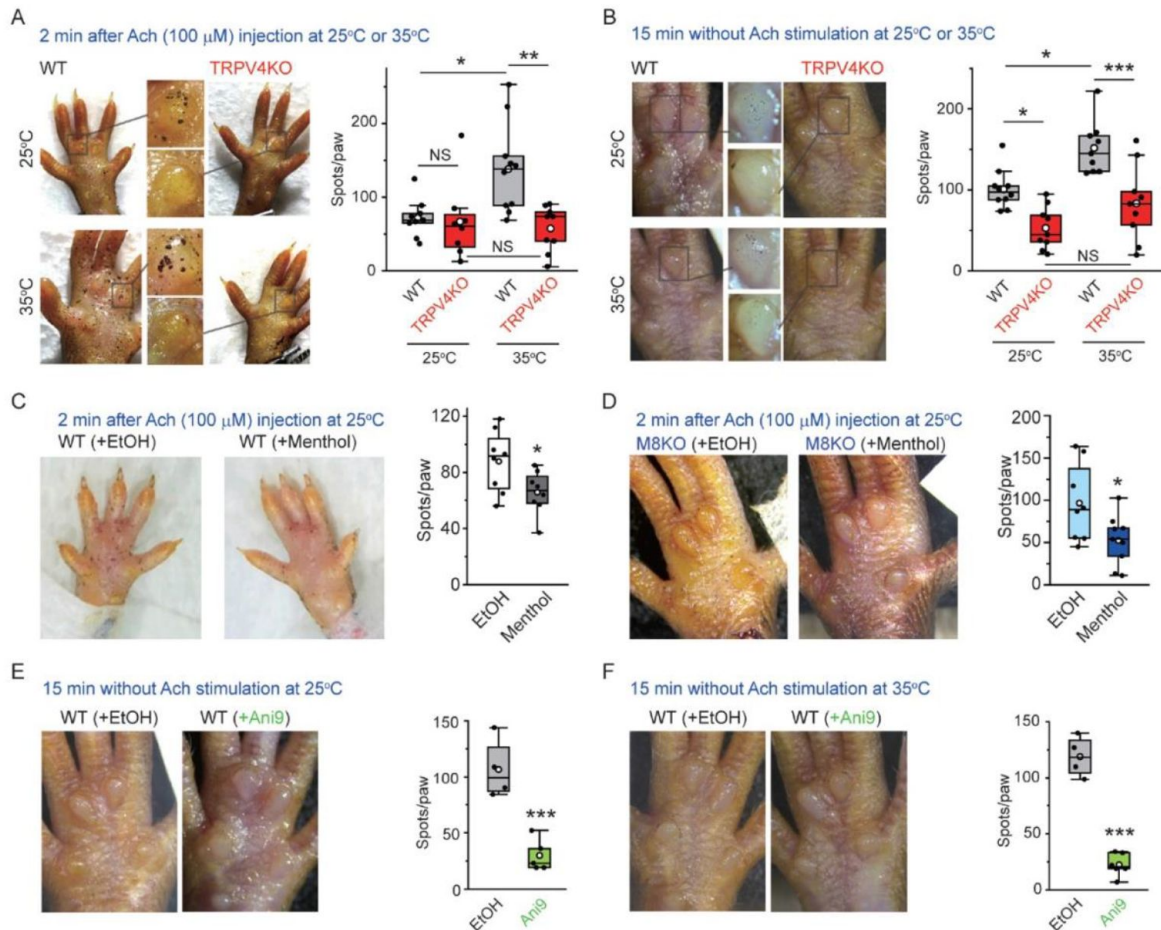


Figure 3.

Stimulated sweating in mouse hind paws at different temperatures.

(A, left) Representative stimulated sweat spots formed at 25 °C or 35 °C in hind paws of WT and TRPV4KO mice 2 min after injection of acetylcholine. (A, right) Comparison of sweat spots/paws at 25 °C or 35 °C in WT and TRPV4KO mice (box-whisker plot). $n = 8\sim 10$ for WT or TRPV4KO. $^{NS}p \geq 0.05$, $^{*}p < 0.05$, $^{**}p < 0.01$ between indicated pairs. (B, left) Representative sweat spots at 25 °C or 35 °C in hind paws of WT and TRPV4KO mice without acetylcholine stimulation at 15 min. (B, right) Comparison of sweat spots/paw at 25 °C or 35 °C for WT and TRPV4KO mice (box-whisker plot). $n = 9\sim 10$ for WT or TRPV4KO. $^{NS}p \geq 0.05$, $^{*}p < 0.05$, $^{***}p < 0.001$ between indicated pairs. (C and D, left) Representative stimulated sweat spots at 25 °C in hind paws of WT (C) and TRPM8KO (D) mice 2 min after injection of acetylcholine with or without menthol. (C and D, right) The effect of menthol on sweat spots/paw at 25 °C in WT (C) and TRPM8KO (D) mice (box-whisker plot). $n = 8\sim 9$ for with or without menthol. $^{*}p < 0.05$. (D and E, left) Representative sweat spots without acetylcholine stimulation at 25 °C (E) and 35 °C (F) in hind paws of WT mice with or without Ani9. (D and E, right) The effect of Ani9 on sweat spots/paw at 25 °C (E) and 35 °C (F) in WT mice (box-whisker plot). $n = 4\sim 6$ for with or without Ani9. $^{***}p < 0.001$.

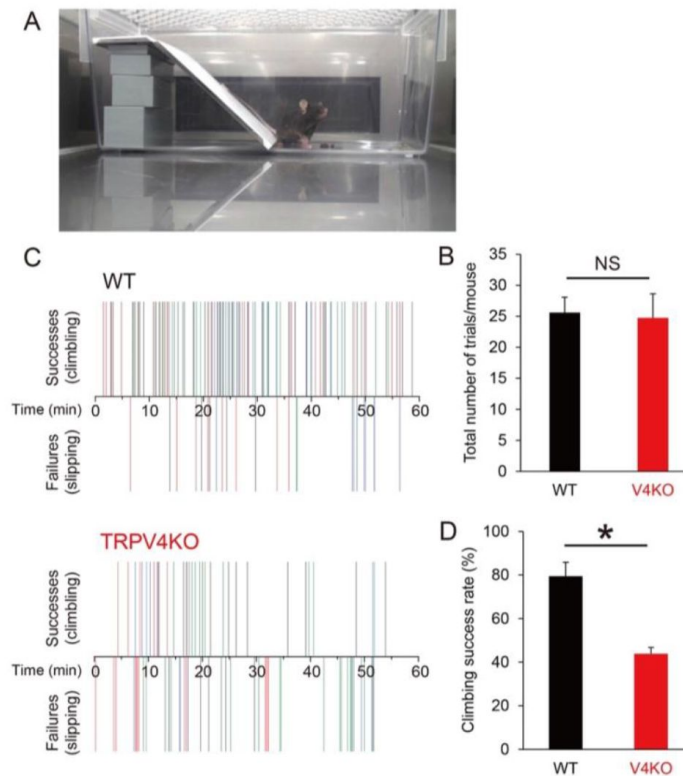


Figure 4.

Climbing behaviors on a slippery slope.

(A) A mouse in a cage containing the vinyl slope. (B) Number of attempts made by WT and TRPV4KO mice within 60 min. (C) Successful (climbing) and failed (slipping) climbing behaviors exhibited by WT and TRPV4KO mice within 60 min. Different colors indicate individual mice. $n = 5$ for both WT and TRPV4KO. (D) Comparison of climbing success rates between WT and TRPV4KO mice. * $p < 0.05$.

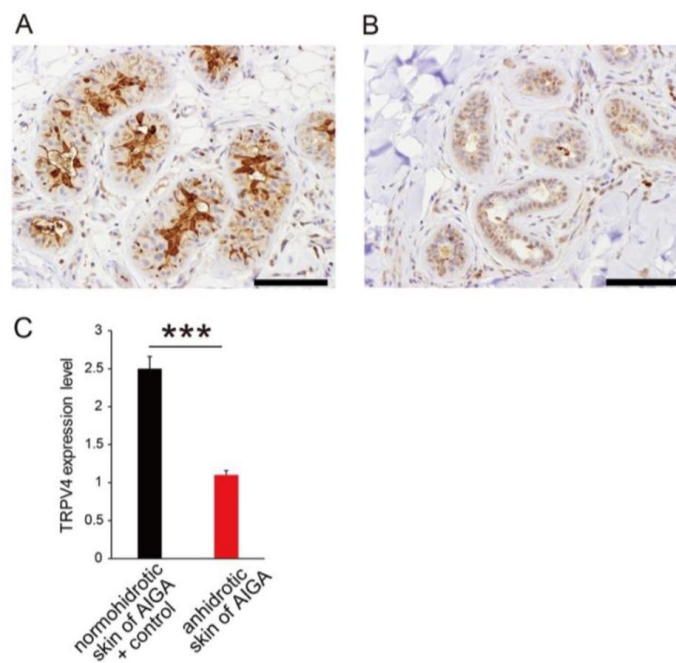


Figure 5.

TRPV4 expression in human sweat glands.

Representative TRPV4 staining in sweat glands from normohidrotic (A) and anhidrotic (B) skin from the same patient with AIGA. Scales bars: 50 µm. (C) Scored TRPV4 expression levels in normohidrotic skin from patients with AIGA and Controls (melanocytic nevus) versus anhidrotic skin from patients with AIGA. *** $p < 0.001$.

efflux, followed by water efflux, suggesting that the complex could be involved in exocrine gland functions including secretion of cerebrospinal fluid, saliva and tears (Takayama, Shibasaki et al. 2014 [DOI](#), Derouiche, Takayama et al. 2018 [DOI](#)). We demonstrated that the TRPV4-ANO1 interaction is also involved in water efflux associated with the exocrine function during sweating in this study. Notably, the TRPV4, ANO1 and AQP5 complex is confined to acinar cells in secretory sweat glands, whereas TRPV4 is also expressed at other sites in skin tissues (**Figure 1** [DOI](#)). This result could indicate a specific function for the complex in water efflux occurring in exocrine glands.

It is generally believed that signals in the brain activates sympathetic nerves which causes perspiration by releasing acetylcholine at the sympathetic postganglionic fibers. This is a simple mechanism for body temperature regulation with perspiration. However, sweat glands themselves could sense local heating, and cause sweating through warmth-sensitive TRPV4 channel activation which we clarified in this study. This local temperature sensation by TRPV4 could be also true for saliva and tear secretion which involves TRPV4/ANO1 interaction. (Derouiche 2018 [DOI](#))

Several human diseases involve hypohidrosis or hyperhidrosis (Cohen and Solish 2003 [DOI](#), Schlereth, Dieterich et al. 2009 [DOI](#), Cheshire 2020 [DOI](#)). Patients with hypohidrosis have difficulty regulating body temperature in response to high temperatures and can experience dizziness, muscle cramps, weakness, high fever or nausea that is typically not serious. However, patients with hypohidrosis sometimes have heatstroke, which is the most serious complication; the incidence of heatstroke has recently increased with global warming. Furthermore, some patients with collagen diseases like Sjögren's syndrome, an autoimmune exocrinopathy, suffer from hypohidrosis as well as dry mouth and dry eye that is not easily treated (Katayama 2018 [DOI](#)). AIGA is also characterized by hypohidrosis without clear etiology (Nakazato, Tamura et al. 2004 [DOI](#), Munetsugu, Fujimoto et al. 2017 [DOI](#), Sano, Asahina et al. 2017 [DOI](#)). In Japan, both Sjögren's syndrome and AIGA are classified as designated intractable diseases (No. 53 and 163, respectively). Problems with exocrine gland function in Sjögren's syndrome patients as well as the low TRPV4 expression levels in patients with AIGA suggest that TRPV4 could be a key molecule involved in these diseases and that novel treatment strategies could target TRPV4 and/or ANO1. There are two recent reports showing that TRPV4 was not critical in regulating sweating in human subjects (Fujii, Kenny et al. 2019 [DOI](#), Fujii, Kenny et al. 2021 [DOI](#)), which is in contradiction to our mouse and human data. The authors focused on the vasodilating effects of TRPV4. We have currently no idea explaining the apparently different conclusion regarding the involvement of TRPV4. Multiple factors could explain the apparent difference between the two studies. The parameters which they examined are different from ours, and the authors examined human healthy volunteers while we used the sample of AIGA patients. More investigation would be needed in the future.

The application of menthol to the skin produces a cool comfortable sensation that is generally thought to result from the activation of the menthol receptor TRPM8. However, the finding here that menthol inhibits both TRPV4 and ANO1 suggests that transient reduction in sweating by inhibiting TRPV4 and ANO1 also contributes to the cool sensation. On the other hand, patients with hyperhidrosis can sweat enough to soak their clothing or have sweat drip off their hands (Cohen and Solish 2003 [DOI](#), Schlereth, Dieterich et al. 2009 [DOI](#)). Hyperhidrosis can occur as a primary or secondary effect after infections or with some endocrine diseases. Others can experience hyperhidrosis on the palms of their hands when nervous. Development of chemicals targeting TRPV4, ANO1 or the complex could be a new therapeutic strategy for these conditions, for which there are currently no effective treatments.

Many TRP channels have high Ca^{2+} permeability, suggesting that Ca^{2+} entering cells through TRP channels in turn activates more Ca^{2+} -activated proteins including other Ca^{2+} -activated ion channels such as Ca^{2+} -activated K^+ channels. This interaction could expand the importance of TRP channels in physiological functions, and complexes between TRP channels and Ca^{2+} -activated proteins would be novel targets for drug development.

Materials and methods

Mice

Homozygous TRPV4-deficient (TRPV4KO) mice from Makoto Suzuki (Jichi Medical University) (Mizuno, Matsumoto et al. 2003 [↗](#)) and homozygous TRPM8-deficient (TRPM8) from Dr. David Julius were maintained under SPF conditions in a controlled environment (12-hour light/dark cycle with free access to food and water, 25 °C, and 50-60% humidity). All procedures were approved by the Institutional Animal Care and Use Committee of the National Institute of Natural Sciences (Approval No. 21A008) and carried out according to the National Institutes of Health and National Institute for Physiological Sciences guidelines. Because it was hard to prepare the enough amount of TRPV4KO mice, we used a little wider distribution of mice age in the immunostaining (8-to 21-week-old mice), and did not use littermates. However, we back crossed the KO mice to the commercially available WT mice more than ten times, causing no difference in the genetic background between WT and TRPV4KO mice.

Human ethics

Informed consent was obtained from all patients, and the study was approved by the Shinshu University Ethics Committee (Approval No. 4073). Anhidrotic or hypohidrotic as well as normohidrotic skin samples taken from various sites were collected from 10 patients who were clinically diagnosed with acquired idiopathic generalized anhidrosis (AIGA) using standard criteria set by the Japan AIGA study group (Revised guideline for the diagnosis and treatment of acquired idiopathic generalized anhidrosis in Japan) (Munetsugu, Fujimoto et al. 2017 [↗](#)).

Chemicals

Collagenase A, trypsin from soybean, ionomycin calcium salt, acetylcholine, carbachol, GSK1016790A (G0798), Ani9 and menthol were purchased from Sigma (St. Louis, MO, USA).

Isolation of sweat glands from mice

Dissected tips of digits and foot pads of mice were minced and incubated in 0.25mg/mL liberase TL (Roche, 5401119001) for 45 min at 37 °C with pipetting every 10min. The digested tissue suspension was filtered through a 40mm cell strainer, and the isolated sweat glands were retained in the filter. The collected sweat glands were seeded on Cell-Tak-coated glass slips and used for Ca²⁺-imaging analysis after incubation at 37 °C (>2h) in DMEM supplemented with 10% fetal bovine serum, penicillin-streptomycin and Glutamax.

Mouse immunostaining

Experiments were performed using 8-to 21-wk-old male and female mice. Mice (n = 4 per group) were anesthetized with a combination of hydrochloric acid medetomidine (0.75mg/kg; Kyoritsu Seiyaku, Tokyo, Japan), butorphanol (5mg/kg; Meiji Seika Pharma, Tokyo, Japan), and midazolam (4mg/kg, Maruishi, 21-3981), and perfused transcardially with heparinized PBS followed by 4% paraformaldehyde in phosphate buffer (pH 7.4). The hind paw pad skin was dissected and post-fixed in 4% PFA for 3 h at 4 °C, cryoprotected with 20% sucrose overnight and then embedded in OCT compound. For immunohistochemistry, 5µm-thick frozen sections were made with a NX50 cryostat. Sections were permeabilized with 0.3% Triton-X100 in PBS for 10 min at room temperature and then incubated with a blocking solution, PBS supplemented with 0.3% Triton X-100, 1% bovine serum albumin, 0.05% sodium azide and 5% normal donkey serum for 45 min at room temperature. Sections were then incubated overnight at 4 °C with the primary antibodies: guinea pig anti-TRPV4 (2 µg/mL) (Kitsuki et al., 2020), rabbit anti-ANO1 (1:100, Abcam, ab53213), rabbit anti-AQP5 (1:200, Millipore, 178615), rabbit anti-CGRP (1:2000, Amersham International,

RPN.1842), rat anti-cytokeratin 8 (CK8) (1:100, Millipore, MABT329), and rat anti-E-cadherin (1:200, Takara Bio, M108). Next, sections were incubated for 1h at room temperature with the secondary antibodies: Alexa Fluor 488 donkey anti-guinea pig IgG, Alexa Fluor 555 donkey anti-rabbit IgG, Alexa Fluor 647 donkey anti-rat IgG (all 1:200, Jackson ImmunoResearch Labs). F-actin was visualized with Phalloidin-iFluor 647 Reagent (1:1000). After immunostaining, sections were incubated for 5 min with 4',6-diamidino-2-phenylindole, dihydrochloride (DAPI, Dojindo, D523) and then mounted with PermaFluor (Thermo Fisher Scientific). Images were acquired using a BC43 or LSM800 instrument equipped with a Zeiss Axio Observer Z1 and a LSM 800 confocal unit with Airyscan module. For super-resolution imaging, images of optical 160 nm-thick slices were taken with a Plan Apochromat 63×/1.40 NA Oil DIC M27 objective. Images were processed with Airyscan processing in ZEN blue 3.5 software.

Human immunostaining

Immunohistochemical analysis of formalin-fixed paraffin embedded tissue sections (2-3 μm-thick) of anhidrotic and normohidrotic skin samples from 10 patients with AIGA. Except for application of primary antibody (100×) all steps including deparaffinization, blocking of internal peroxidase activity, unmasking of specific antigen, application of secondary antibody, detection of signals and nuclear staining were automatically performed by a Ventana auto-staining system. Skin samples with melanocytic nevus (n=10) were used as a control.

Iodine and starch test

Iodine starch assay was conducted using wild-type, TRPV4KO, and TRPM8KO male mice (8-14 weeks old) under room temperature (25°C) or 35°C at 30-60% humidity. Mice were anesthetized by isoflurane inhalation and held in a prone position. Soles of hind paw were painted with 3% iodine/EtOH and stand for 2min to dry solvent. Thereafter, same surfaces were painted with 10% starch/mineral oil, and positive signals of iodine-starch reaction (dark-blue color) were counted after 15min for sweating without acetylcholine stimulation. For the experiments with acetylcholine injection, acetylcholine (100 μM) in PBS (-) was subcutaneously administered after the starch painting and the signals were counted after 2min after acetylcholine injection. For the analyses of Ani9 and menthol, Ani9 (10 μM in EtOH) or menthol (0.5% in EtOH) was topically applied to the soles 10 min before iodine-starch test.

Calcium imaging

After loading with Fura-2 AM (5 μM, Invitrogen, F-1201), isolated sweat glands on coverslips were mounted in an open chamber and rinsed with standard bath solution containing (in mM): 140 NaCl, 5 KCl, 2 MgCl₂, 2 CaCl₂, 10 HEPES, and 10 glucose, pH 7.4. The intracellular free calcium concentration in isolated sweat glands was measured by dual-wavelength fura-2 microfluorometry with excitation at 340/380 nm and emission at 510 nm. The ratio image was calculated and acquired using the IP-Lab imaging processing system.

Mouse climbing experiments

WT and TRPV4KO mice were allowed to acclimate for one day prior to recording in a cage containing a slippery slope made with vinyl. Mice were housed for 1 hour in the cage with the slope at 26-27 °C and 35-50% humidity. Climbing and slipping behaviors were videotaped and analyzed.

Quantifications and statistical analysis

Data are shown as mean ± sem. Statistical analysis was performed with Origin Pro 8. Student's t-test and two-way ANOVA with Dunnett's or Bonferroni's multiple-comparison tests were performed for comparisons. Values of p <0.05 indicate statistical significance.

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Author contribution

M.K., S.D, R.U.Y., K.S., J.L. and M.A.K. designed and performed the experiments. M.A.K. and M.T. wrote the paper.

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Joint Public Review:

In this study, Kashio et al examined the role of TRPV4 in regulating perspiration in mice. They find coexpression of TRPV4 with the chloride channel ANO1 and aquaporin 5, which implies possible coupling of heat sensing through TRPV4 to ion and water excretion through the latter channels. Calcium imaging of eccrine gland cells revealed that the TRPV4 agonist GSK101 activates these cells in WT mice, but not in TRPV4 KO. This effect is reduced with

cold-stimulating menthol treatment. Temperature-dependent perspiration in mouse skin, either with passive heating or with ACh stimulation, was reduced in TRPV4 KO mice. Functional studies in mice - correlating the ability to climb a slippery slope to properly regulate skin moisture levels - reveal potential dysregulation of foot pad perspiration in TRPV4 KO mice, which had fewer successful climbing attempts. Lastly, a correlation of TRPV4 to hypohydrosis in humans was shown, as anhidrotic skin showed reduced levels of TRPV4 expression compared to normohidrotic or control skin.

Overall this is an interesting study on how TRPV4 regulates perspiration.

- (1) The functional relationship between TRPV3 and ANO1 remains correlative.
- (2) Littermate controls were not used, but TRPV4ko were backcrossed onto the WT strain.
- (3) In general, the results support the authors' claims that TRPV4 activity is a necessary component of sweat gland secretion, which may have important implications for controlling perspiration; secretion from other glands where TRPV4 may be expressed remains a possibility given the lack of us of exocrine-specific knockouts.

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Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

(1) Measurement of secreted amylase could be seen as direct evidence of sweating, however, how to determine the causal relationship between climbing behavior and sweating? Friction force may also be reduced when there is too much fingertip moisture.

As the reviewer notes, measurement of secreted amylase can provide direct evidence of sweating, and we performed an iodine and starch reaction. Upon observing the involvement of TRPV4 in mouse foot pad perspiration, we then considered which type of behavioral analysis would be suitable to evaluate this perspiration. We agree with the reviewer's point that friction force in the climbing test may be reduced by excessive sweating. However, we did not observe severe sweating in the absence of acetylcholine treatment. Accordingly, we interpreted that the increase in the climbing test failure rate for TRPV4KO mice could reflect the reduced friction force associated with the lack of TRPV4 activity.

(2) For the human skin immunostaining, did the author use the same TRPV4 antibody as used in the mouse staining? Did they validate the specificity of the antibody for the human TRPV4 channel?

We used different antibodies for human and mouse samples. Since commercially available anti-TRPV4 antibodies do not work well with mouse samples, we generated our own anti-TRPV4 antibody and validated its specificity.

(3) In lines 116-117, the authors tried to determine "the functional interaction of TRPV4 and ANO1 is involved in temperature-dependent sweating", however, they only used the TRPV4 ko mice and did not show any evidence supporting the relationship between TRPV4 and ANO1.

As the reviewer pointed out, based on the data presented in the original submission we cannot conclude that an interaction between TRPV4 and ANO1 is involved in perspiration.

However, we think that the data for TRPV4KO mice presented in Figure 3 of the original version does indicate that TRPV4 is involved in perspiration. The finding that menthol and its related compounds, which inhibit the function of both TRPV4 and ANO1 (see our publication in *Scientific Reports* 7: 43132, 2017), blocked perspiration in both wild-type and TRPV4KO mice (original Figure 3C, D) indicates involvement of either TRPV4 or ANO1 in perspiration. In the revised version, we present results for additional iodine and starch reaction experiments using Ani9, a potent and specific ANO1 inhibitor. Ani9 drastically inhibited perspiration from mouse food pads both at 25 °C and 35 °C. Based on these collective results, we concluded that both TRPV4 and ANO1, likely acting as a complex, are involved in perspiration. We present the new data with Ani9 in the revised Figure 3E, F.

(4) Figure 3-4 is quite confusing. At 25°C, no sweating difference was observed between TRPV4 and wt mice (Fig 3A-3D), suggesting both Ach-induced sweating and basal sweating are TRPV4-independent at 25°C, however, the climbing test was done at 26-27 °C and the data showed a climbing deficit in TRPV4 ko mice. How to interpret the data is unclear.

Thank you for raising this point. In the iodine and starch reaction experiment, we observed no significant reduction in perspiration in the absence of acetylcholine at 25 °C, which is the same condition as in the climbing test, whereas we detected less perspiration for TRPV4KO mice. In a trial using additional mice, we detected significantly less perspiration under control conditions without acetylcholine at 25 °C, which is consistent with the results of the climbing test. We have added this new data to the revised Figure 3A, B.

(5) Were there any gender differences associated with sweating in mice? In Figure 3, the mouse number for behavior tests should be at least 5.

The TRPV4KO mice reproduced poorly and we were unable to obtain sufficient numbers of male and female mice to determine whether there were gender differences in sweating. However, according to the reviewer's suggestion, and as mentioned above, we increased the number of experiments to obtain the results shown in the revised Figure 3. We did not observe a significant difference in sweating with the larger sample size, which supports our conclusions.

(6) 8- to 21-week-old mice were used in the immunostaining, the time span is too long.

Given the difficulty in obtaining sufficient numbers of TRPV4KO mice, we used a somewhat wider age distribution to obtain samples for immunostaining. However, we did not observe age-dependent differences in immunostaining. We reference this point in the revised manuscript.

(7) The authors used homozygous TRPV4 ko mice for all experiments. What are control mice? Are they littermates of the TRPV4 ko mice?

We did not use littermates for our in vivo experiments because the TRPV4KO mice reproduced poorly and the litter sizes were small. However, we did backcross the KO mice to the commercially available wild-type mice more than ten times. As such, we expect that the wild-type and TRPV4KO mice will have similar genetic backgrounds. In addition, we have published multiple studies that have successfully used this method, which we think supports the reliability of our results for experiments involving mice.

Reviewer #2 (Public Review):

(1) The coexpression data needs additional controls. In the TRPV4 KO mice, there appears to be staining with the TRPV4 Ab in TRPV4 KO mice below the epidermis. This pattern appears similar to that of the location of the secretory coils of the sweat glands (Fig 1A). Is the co-staining the authors note later in Figure 1 also seen in TRPV4 KOs? This control should be shown, since the KO staining is not convincing that the Ab doesn't have off-target binding.

We thank the reviewer for raising these concerns about immunostaining. As the reviewer notes, in the low power image the signals appeared to be weak and punctate signals were present in the basal region of glandular cells. Although we did not identify immunohistochemical conditions that produced no signal, tissue sections from WT mice stained with anti-TRPV4 antibody showed conspicuous apical signals for the glandular cells facing lumen. Meanwhile, TRPV4KO tissues showed no signals at the apical region of the glandular cells, where the TRPV4-ANO1 interaction is expected to occur. We confirmed no trace signals in the TRPV4KO tissues in the immunoblotting.

(2) Are there any other markers besides CGRP for dark cells in mice to support the conclusion that mouse secretory cells have clear cell and dark cell properties?

We did not stain with other dark cell markers. Based on previous studies describing the differences between clear and dark cells in mouse eccrine glands, we think that dark and clear cells cannot be clearly discriminated, as we described in lines 93-96 of the Results. We identified secretory cells using CK8 and dark cells with CGRP, a marker of dark cells in human eccrine glands (Zancanaro et al. 1999 J Anat). Our result showed that CGRP immunostaining could not discriminate between clear and dark cells, which is consistent with a previous report showing that mouse secretory cells were assumed to be undifferentiated and primitive based on electron microscopic observation (Kurosumi et al. 1970 Arch Histol Jap).

(3) The authors utilize menthol (as a cooling stimulus) in several experiments. In the discussion, they interpret the effect of menthol as potentially disrupting TRPV4-ANO1 interactions independent of TRPM8. Yet, the role of TRPM8, such as in TRPM8 KO mice, is not evaluated in this study.

We performed the iodine and starch reaction experiments with TRPM8KO mice. In the TRPM8KO mice, the sweat spots did not differ from those seen for WT mice ($p=0.63$, t-test), and there was also a significant reduction in sweating with menthol treatment following acetylcholine stimulation that was similar to that seen for WT mice. These results would rule out the involvement of TRPM8 in a menthol-induced reduction in sweating. We have included this data in the revised Figure 3D.

(4) Along those lines, the authors suggest that menthol inhibits eccrine function, which might lead to a cooling sensation. But isn't the cooling sensation of sweating from evaporative cooling? In which case, inhibiting eccrine function may actually impair cooling sensations.

Menthol has a non-specific effect that activates TRPM8, TRPV3 and TRPA1, and inhibits TRPV1, TRPV4 and ANO1. Therefore, we did not carry out a climbing test with menthol in part because menthol-dependent TRPA1 activation decreased the propensity of the mice to climb. As the reviewer notes, TRPM8 activation following topical application of menthol may cause a cooling sensation elicited in sensory neurons beneath the skin. However, the comfortable cooling sensation could also be caused in part by decreased sweating. The relationship

between a comfortable cooling sensation and less perspiration following menthol application may be difficult to determine, and we have mentioned this in the updated Discussion.

(5) The climbing assay is interesting and compelling. The authors note performing this under certain temperature and humidity conditions. Presumably, there is an optimal level of skin moisture, where skin that is too dry has less traction, but skin that is too wet may also have less traction. It would bolster this section of the study to perform this assay under hot conditions (perhaps TRPV4 KO mice, with impaired perspiration, would outperform WT mice with too much sweating?), or with pharmacologic intervention using TRPV4 agonists or antagonists to more rigorously evaluate whether this model correlates to TRPV4 function in the setting of different levels of perspiration.

We thank the reviewer for this suggestion. Upon detecting the involvement of TRPV4/ANO1 interaction in perspiration, we considered different behavioral analyses that can be performed to demonstrate whether the TRPV4/ANO1 interactions are involved in perspiration. As the reviewer suggested, there should be an optimal level of sweating. Therefore, we first set the room temperature at 26-27 °C and humidity at 35-50%. To our knowledge, this is the first demonstration of temperature-dependent sweating of mouse foot pads. In humans, palm sweating is often referred to as psychotic sweating that is known to be regulated by sympathetic nerve activity. Here we tested whether foot pad sweating might be related to friction force wherein sufficient amounts of sweating could increase the friction force and in turn increase the success rate for the climbing test using a vinyl-covered slippery slope that was selected based on several trials to determine the optimal surface material and slope angles. As the reviewer suggests, the success rates could be affected by multiple factors, and hot temperatures likely induce more sweating that could increase the success rates in the climbing test. We will need to carry out additional experiments that are beyond the scope of this study to examine these temperature-dependent effects. Generally, sweating is regulated by sympathetic nerve activity that occurs in response to increased brain neuron excitation. However, here we raise for the first time the possibility that sweating might be regulated by local temperature sensation mediated through TRPV4 that may be effective for fine-tuning of perspiration activity. We have updated the Discussion to reference this possibility.

(6) There are other studies (PMID 33085914, PMID 31216445) that have examined the role of TRPV4 in regulating perspiration. The presence of TRPV4 in eccrine glands is not a novel finding. Moreover, these studies noted that TRPV4 was not critical in regulating sweating in human subjects. These prior studies are in contradiction to the mouse data and the correlation to human anhidrotic skin in the present study. Neither of these studies is cited or discussed by the authors, but they should be.

We thank the reviewer for referencing these other studies concerning the possible involvement of TRPV4 in perspiration in humans. These studies focused on the vasodilating effects of TRPV4 and drew the conclusion that TRPV4 is not involved in sweating in humans, which is in contrast to our data for mice and humans. Multiple factors could explain the apparent difference between the two studies. For example, the parameters they examined differed from ours in that we assessed patients with AIGA, whereas the previous studies involved healthy volunteers. We have updated the Discussion to note the difference in the results of our and previous studies.

Reviewer #3 (Public Review):

(1) Figure 2: The calcium imaging-based approach shows average traces from 6 cells per genotype, but it was unclear if all acinar cells tested with this technique demonstrated TRPV4-mediated calcium influx, or if only a subset was presented.

“n = 6” does not indicate the number of cells, but rather 6 independent experiments that each had over 20 ROIs of sweat glands. We have clarified this point in the updated figure legend.

(2) Figure 4: The climbing behavioral test shows a significant reduction in climbing success rate in TRPV4-deficient mice. The authors ascribe this to a lack of hind paw 'traction' due to deficiencies in hind paw perspiration, but important controls and evidence that could rule out other potential confounds were not provided or cited.

As noted in our response to Comment 5 made by Reviewer #2, we spent considerable time identifying optimal conditions that would delineate success rates in the climbing experiments. We are confident that TRPV4KO mice had significantly lower success rates than WT mice, but there are various factors that could affect the experimental outcomes. We reference these factors in the updated Discussion.

(3) In general, the results support the authors' claims that TRPV4 activity is a necessary component of sweat gland secretion, which may have important implications for controlling perspiration as well as secretion from other glands where TRPV4 may be expressed.

As described above, the results we obtained in the climbing test can be affected by various factors. However, based on the consistency of the results obtained for the climbing test and the iodine and starch reaction assay, we think that our interpretation is correct. In terms of the involvement of TRPV4/ANO1 interactions in fluid secretion, we previously reported that the TRPV4/ANO1 complex is involved in cerebrospinal fluid secretion in the mouse choroid plexus (FASEB J. 2014) and in saliva and tear secretion in mouse salivary and lacrimal glands (FASEB J. 2018). Together, these findings suggest that this mechanism is common to water efflux from exocrine glands.

Reviewer #1 (Recommendations For The Authors):

(1) An exocrine gland-specific trpv4 knockout mouse should be used, as TRPV4 is also expressed by muscles, global knockout TRPV4 may affect the TRPV4-dependent muscle strength and reduce the climbing ability in mice.

As the reviewer suggests, use of mice with TRPV4 knockout specific to exocrine glands would be preferable to mice having global TRPV4 knockout given that TRPV4 is expressed in multiple tissues. We agree with this suggestion, but we do not currently have such mice in hand. However, as mentioned above, we have reported the involvement of the TRPV4/ANO1 interaction in cerebrospinal fluid secretion from the choroid plexus in mice (FASEB J. 28: 2238-2248, 2014), as well as saliva and tear secretion in mouse salivary and lacrimal glands (FASEB J. 32: 1841-1854, 2018.), suggesting that the TRPV4/ANO1 interaction could be widely involved in exocrine gland functions that involve water movement. We have updated the Discussion to reference this point.

(2) The authors showed Calcium imaging data that Menthol inhibits TRPV4-dependent calcium influx. However, it is well known that menthol induces the sensation of cooling by activating TRPM8. More evidence, including patch clamp recordings, should be done to verify the inhibition effects of menthol on TRPV4 and ANO1. Moreover, Fig 3E-3F could only suggest that menthol-induced cooling sensation may affect sweating but not the inhibition effect of menthol on TRPV4 and ANO1 channels.

We agree that more evidence including patch-clamp recordings can verify the inhibitory effects of menthol on TRPV4 and ANO1. We did not include such experiments here since we

previously showed that menthol and related agents indeed inhibit TRPV4- and ANO1-mediated currents (Sci. Rep. 7: 43132, 2017). We now cite this paper in the revised version.

(3) Excepting the climbing test, are there any other better models to assess the sweating-related behaviors?

When we detected the involvement of TRPV4/ANO1 interactions in perspiration, we considered different types of behavioral analyses that could be used to demonstrate TRPV4/ANO1-dependent perspiration. We think that the climbing experiment is the best test, particularly since foot pads are one of the few regions on mice that is not covered by fur and thus amenable to evaluation of perspiration using an iodine and starch test.

Reviewer #2 (Recommendations For The Authors):

(1) I was confused by a section in the introduction on lines 59-60: How does Cl⁻ efflux lead to the formation of a physical complex in cells with high intracellular Cl⁻? What is the physical complex? This seems like several disparate concepts combined together, which need to be clarified.

We apologize for the incomplete descriptions of several of our previous works. We have amended the Introduction section in the revised manuscript.

Reviewer #3 (Recommendations For The Authors):

(1) TRPV4 is expressed by multiple other cell types in the skin (keratinocytes, macrophages etc.) which may have an impact on peripheral sensory function. Is there evidence that TRPV4-deficient animals have relatively normal sensory acuity and/or proprioception? Such evidence would lend more credibility to the reported findings in the climbing test.

As the reviewer points out, TRPV4 is expressed by multiple other cell types in the skin. To date we have found that TRPV4KO mice show no differences in sensory functions compared to WT mice. Whether TRPV4 is involved in proprioception is unclear, based on both our own observation and those that appear in the literature, although TRPV4 is clearly activated by mechanical stimuli. We previously compared the mechanical sensitivity of TRPV4 and Piezo1 in bladder epithelial cells, and found that Piezo 1 shows much higher sensitivity relative to TRPV4 (J. Biol. Chem. 289: 16565-16575, 2014), which is consistent with the involvement of Piezo1, rather than TRPV4, in proprioception. Although TRPV4 is reported to be expressed in sensory neurons, we did not detect TRPV4-mediated responses in isolated rat and mouse DRG neurons, suggesting that TRPV4-positive sensory neurons are relatively rare.

(2) The methods section refers to loading entire sweat glands with Fura-2 dye for calcium imaging, but the figure legend refers to sweat gland acinar cells. Resolving this ambiguity would help readers to interpret the data.

We apologize for this error and have made an appropriate correction in the revised manuscript.

(3) Alternatively, could acute intraplantar injection of a TRPV4 antagonist (e.g. GSK205) in wild-type mice phenocopy the TRPV4-knockout mouse deficits, or could normal climbing behavior be restored in the TRPV4 knockout by adding artificial perspiration to their hindpaws?

We thank the reviewer for raising this interesting possibility and suggesting use of TRPV4 agonists or antagonists in the climbing tests. We agree that results of such an experiment

would support the involvement of TRPV4 in sweating. We tried to do such experiments using injection of TRPV4 regulators into mouse hindpaws. However, the injections themselves appeared to impact climbing ability, perhaps in part due to painful sensations associated with the injection. Similarly, menthol injection appeared to reduce climbing activity, likely through pain sensations associated with TRPA1 activation. As such, we did not pursue these experiments.

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